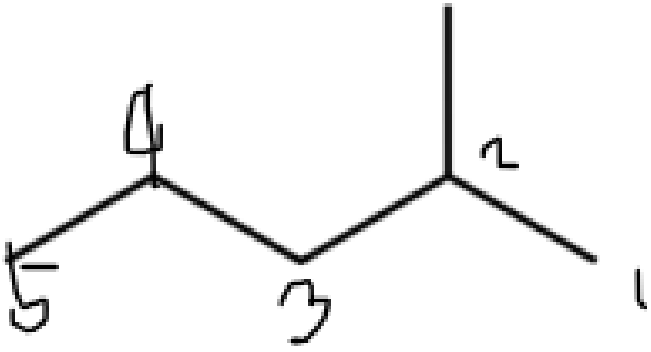
CARD A01

MOLECULAR FORMULA:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted



NUMBERED PARENT

PARENT: LONGEST CHAIN = 5C > PENTANE. 2 PATHS EXIST, BUT STILL END UP WITH ONE METHYL, THEREFORE SAME NAME

NUMBERING: L>R GIVES 4, R>L GIVES 2. FIRST PT DIFFERENCE: 2<4, SO NUMBERED FROM THE RIGHT.

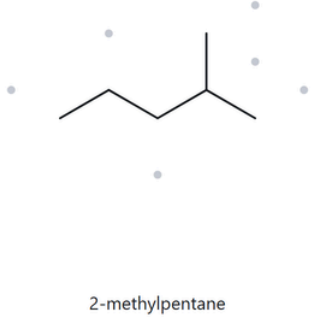
SUBSTITUENTS: METHYL ON C2> 2-METHYL.

NAMING TRAIL

2-METHYLPENTANE

MY NAME (WRITTEN FIRST)

Typed before the tool was opened



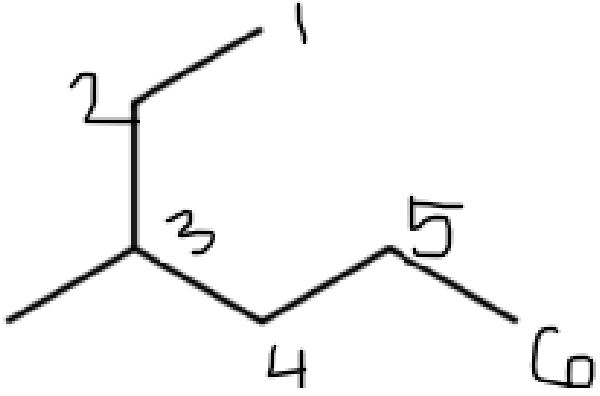
THE ALKANE CHECK

Result name: 2-methylpentane
A MATCH!

CARD A02

MOLECULAR FORMULA:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted



NUMBERED PARENT

PARENT: LONGEST CHAIN = 6C > HEXANE. ONLY ONE PATH EXISTS TO GET 6C, THEREFORE SAME NAME.

NUMBERING: L>R GIVES 3, R>L GIVES 4. FIRST PT DIFFERENCE: 3<4, SO NUMBERED FROM THE LEFT.

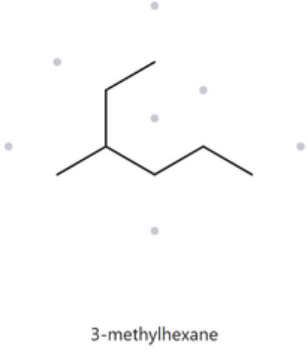
SUBSTITUENTS: METHYL ON 3C> 3-METHYL.

NAMING TRAIL

3-METHYLHEXANE

MY NAME (WRITTEN FIRST)

Typed before the tool was opened



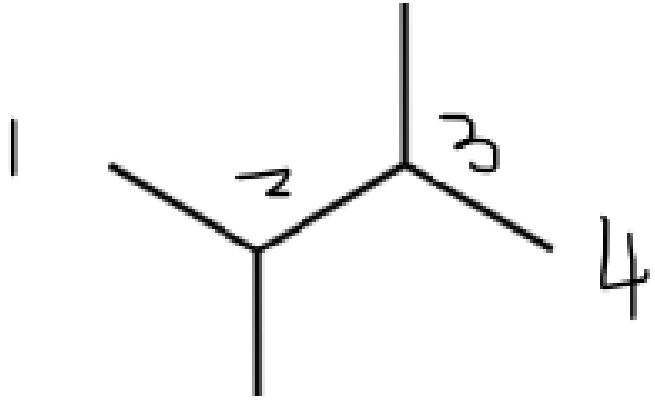
THE ALKANE CHECK

Result name: 3-methylhexane
A MATCH!

CARD A03

MOLECULAR FORMULA:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted



NUMBERED PARENT

PARENT: LONGEST CHAIN = 4C > BUTANE. ALL PATHS LEAD TO 4C, THEREFORE SAME NAME.

NUMBERING: L>R GIVES 2,3, R>L GIVES 2,3. FIRST PT DIFFERENCE: NONE!, SO CAN BE NUMBERED EITHER WAY!

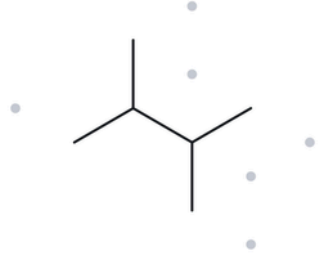
SUBSTITUENTS: METHYL ON 2C, 3C> 2,3-DIMETHYL.

NAMING TRAIL

2,3-DIMETHYLBUTANE

MY NAME (WRITTEN FIRST)

Typed before the tool was opened



2.3-dimethylbutane

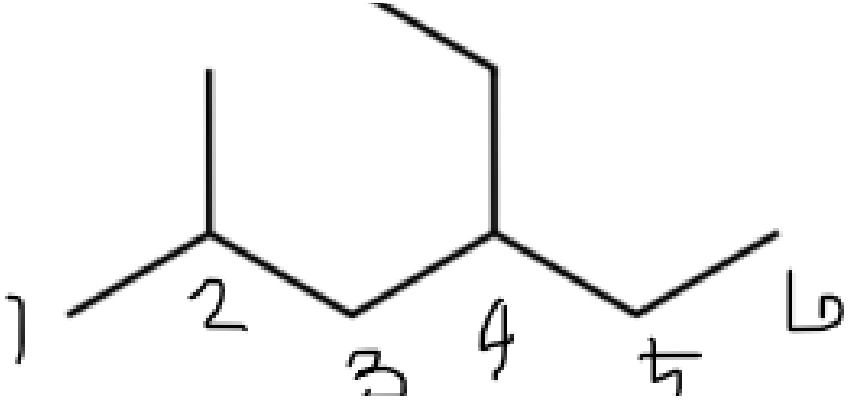
THE ALKANE CHECK

Result name: 2,3-dimethylbutane
A MATCH!

CARD A04

MOLECULAR FORMULA:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted



NUMBERED PARENT

PARENT: LONGEST CHAIN = 6C > HEXANE. ONLY TWO PATHS LEAD TO 6C, THEREFORE SAME NAME.

NUMBERING: L>R GIVES 2, 4, R>L GIVES 3, 5. FIRST PT DIFFERENCE: 2<3, SO NUMBERED FROM THE LEFT

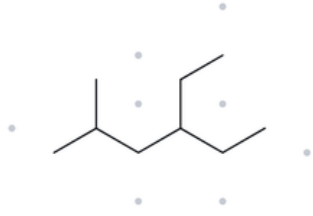
SUBSTITUENTS: ALPHABETICALLY, ETHYL ON 4C, METHYL ON 2C, THEREFORE 4-ETHYL, 2-METHYL

NAMING TRAIL

4-ETHYL, 2-METHYLHEXANE

MY NAME (WRITTEN FIRST)

Typed before the tool was opened



4-ethyl-2-methylhexane

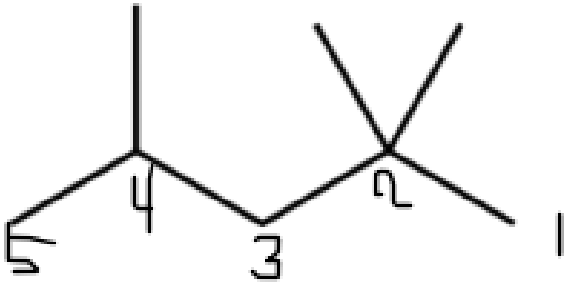
THE ALKANE CHECK

Result name: 4-ethyl, 2-methylhexane
A MATCH!

CARD A05

MOLECULAR FORMULA:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted



NUMBERED PARENT

PARENT: LONGEST CHAIN = 5C > PENTANE. ALL PATHS LEAD TO 5C, THEREFORE SAME NAME.

NUMBERING: L>R GIVES 2, 4, 4 R>L GIVES 2, 2, 4. FIRST PT DIFFERENCE: 2<4, SO NUMBERED FROM THE RIGHT.

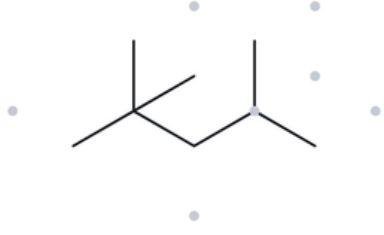
SUBSTITUENTS: 2 METHYL ON 2C, 1 METHYL ON 4C, THEREFORE 2,2,4-TRIMETHYL

NAMING TRAIL

2,2,4-TRIMETHYLPENTANE

MY NAME (WRITTEN FIRST)

Typed before the tool was opened



2,2,4-trimethylpentane

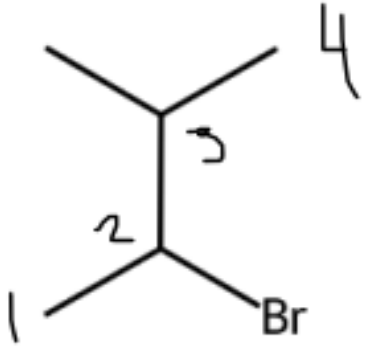
THE ALKANE CHECK

Result name: 2,2,4-trimethylpentane
A MATCH!

CARD A06

MOLECULAR FORMULA:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted



NUMBERED PARENT

PARENT: LONGEST CHAIN = 4C > BUTANE. ALL PATHS LEAD TO 4C, THEREFORE SAME NAME.

NUMBERING: L>R GIVES 2, 3 R>L GIVES 3, 2. FIRST PT DIFFERENCE: 2>3 SO NUMBERED FROM THE LEFT.

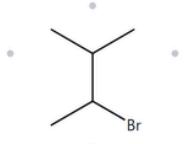
SUBSTITUENTS: ALPHABETICALLY, 1 BROMINE ON 2C, 1 METHYL ON 3C, THEREFORE 2-BROMO-3-METHYL

NAMING TRAIL

2-BROMO-3-METHYLBUTANE

MY NAME (WRITTEN FIRST)

Typed before the tool was opened



2-bromo-3-methylbutane

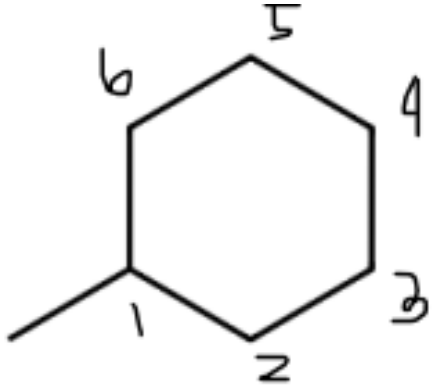
THE ALKANE CHECK

Result name: 2-bromo-3-methylbutane
A MATCH!

CARD A07

MOLECULAR FORMULA:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted



NUMBERED PARENT

PARENT: THE RING HAS MORE C: 6<2. THEREFORE, CYCLOHEXANE IS THE PARENT CHAIN

NUMBERING: L>R GIVES 1, SO NUMBERED FROM THE LEFT.

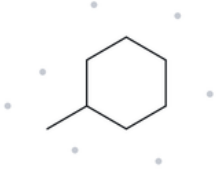
SUBSTITUENTS: 1 METHYL ON 1C, THEREFORE METHYL.

NAMING TRAIL

METHYLCYCLOHEXANE

MY NAME (WRITTEN FIRST)

Typed before the tool was opened



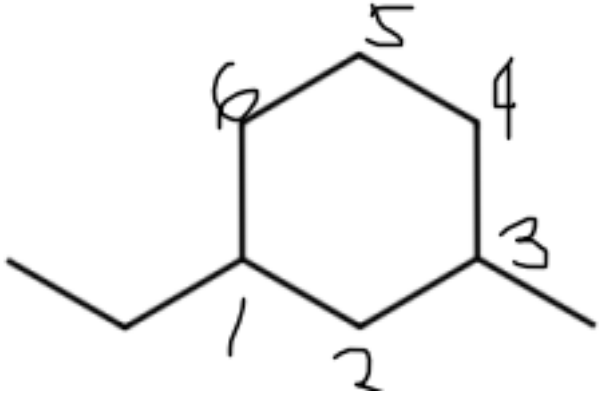
THE ALKANE CHECK

Result name: methylcyclohexane
A MATCH!

CARD A08

MOLECULAR FORMULA:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted



NUMBERED PARENT

PARENT: THE RING HAS MORE C: 6<3. THEREFORE, CYCLOHEXANE IS THE PARENT CHAIN

NUMBERING: L>R GIVES 1, 3, R>L GIVES 3, 1 SO NUMBERED FROM THE LEFT.

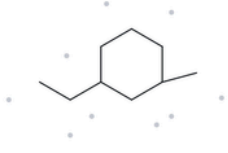
SUBSTITUENTS: ALPHABETICALLY, 1 ETHYL ON 1C, 1 METHYL ON 3C, THEREFORE 1-ETHYL-3-METHYL.

NAMING TRAIL

1-ETHYL-3-METHYLCYCLOHEXANE

MY NAME (WRITTEN FIRST)

Typed before the tool was opened



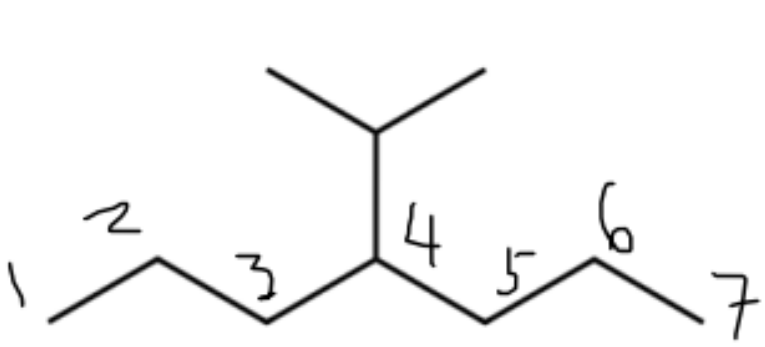
THE ALKANE CHECK

Result name: 1-ethyl-3-methylcyclohexane
A MATCH!

CARD A09

MOLECULAR FORMULA:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted



NUMBERED PARENT

PARENT: LONGEST CHAIN = 7C > HEPTANE. ONLY ONE PATH RESULTS IN 7C, THEREFORE HEPTANE

NUMBERING: L>R GIVES 4, R>L GIVES 4, THEREFORE EITHER WAY.

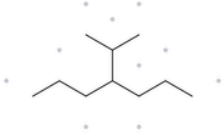
SUBSTITUENTS: 1 ISOPROPYL ON 4C, THEREFORE 4-ISOPROPYL

NAMING TRAIL

4-ISOPROPYLHEPTANE

MY NAME (WRITTEN FIRST)

Typed before the tool was opened



2-methyl-3-propylhexane

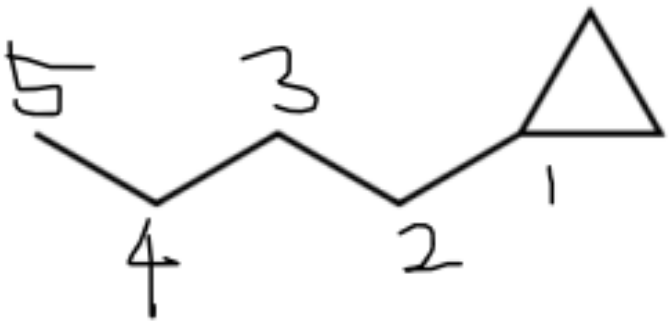
THE ALKANE CHECK

Result name: 2-methyl-3-propylhexane
NOT A MATCH! NOOOoooooo!! (My name is correct, it follows the rule of the longest chain.)

CARD A10

MOLECULAR FORMULA:

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted



NUMBERED PARENT

PARENT: LONGEST CHAIN = 5C > PENTANE. ONLY ONE PATH RESULTS IN 5C, THEREFORE PENTANE

NUMBERING: L>R GIVES 5, R>L GIVES 1, THEREFORE NUMBERED FROM THE RIGHT

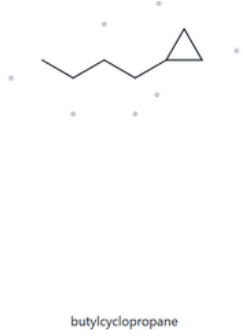
SUBSTITUENTS: 1 CYCLOPROPYL ON 1C, THEREFORE 1-CYCLOPROPYL

NAMING TRAIL

1-CYCLOPROPYLPENTANE

MY NAME (WRITTEN FIRST)

Typed before the tool was opened



THE ALKANE CHECK

Result name: butylcyclopropane
NOT A MATCH! nooo!!! (My name is correct, it follows the rule of the longest chain.)

CARD B01

MOLECULAR FORMULA:

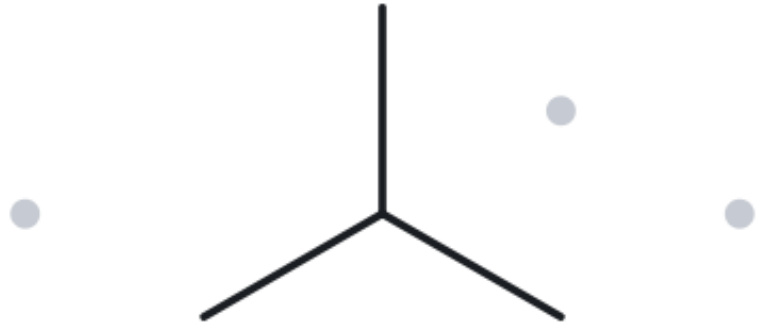
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

PARENT: PROPANE > A THREE CARBON CHAIN

SUBSTITUENTS: METHYL ON C2.

EXPECTED FORMULA: 3+1 = 4C, ONE CHAIN, ONE SUBSTITUENT> C4H10

NAME DECODED



BOND-LINE STRUCTURE (THE ALKANE)

COUNTED FROM THE DRAWING: C4H10

METHYL ON C2

THEREFORE CORRECT!

FORMULA CHECK

RETURNED BY THE ALKANE: 2-METHYLPROPANE

SAME AS GIVEN NAME!

BACK-CHECK

CARD B02

MOLECULAR FORMULA:

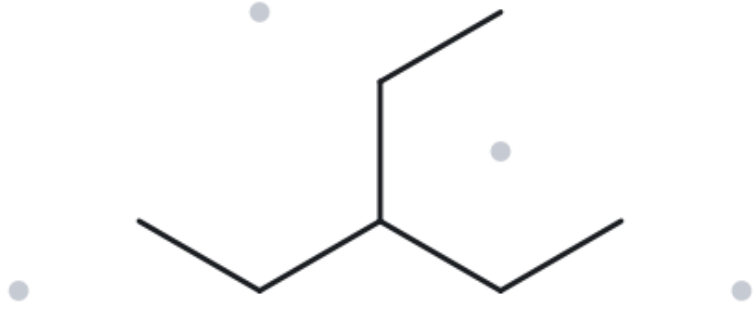
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

PARENT: PENTANE> A FIVE CARBON CHAIN

SUBSTITUENTS: ETHYL ON C3.

EXPECTED FORMULA: 5+2 = 7C, ONE CHAIN, ONE SUBSTITUENT> C7H16

NAME DECODED



BOND-LINE STRUCTURE (THE ALKANE)

COUNTED FROM THE DRAWING: C7H16

ETHYL ON C3

THEREFORE CORRECT!

FORMULA CHECK

RETURNED BY THE ALKANE: 3-ETHYLPENTANE

SAME AS GIVEN NAME!

BACK-CHECK

CARD B03

MOLECULAR FORMULA:

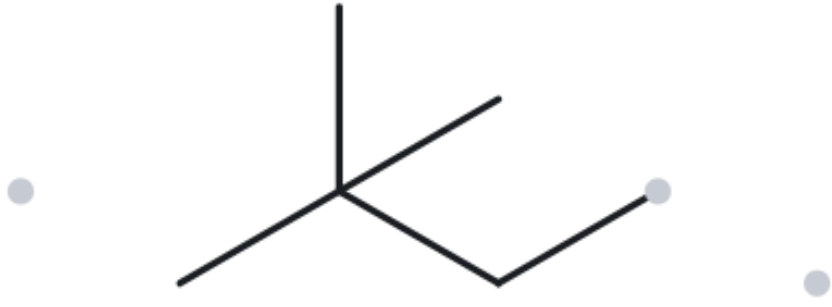
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

PARENT: BUTANE> A FOUR CARBON CHAIN

SUBSTITUENTS: TWO METHYL ON C2 FROM THE LEFT, THEREFORE DIMETHYL

EXPECTED FORMULA: 4+2 = 6C, ONE CHAIN, TWO SUBSTITUENTS> C6H14

NAME DECODED



BOND-LINE STRUCTURE (THE ALKANE)

COUNTED FROM THE DRAWING: C6H14

DIMETHYL ON C2

THEREFORE CORRECT!

FORMULA CHECK

RETURNED BY THE ALKANE: 2,2-DIMETHYLBUTANE

SAME AS GIVEN NAME!

BACK-CHECK

CARD B04

MOLECULAR FORMULA:

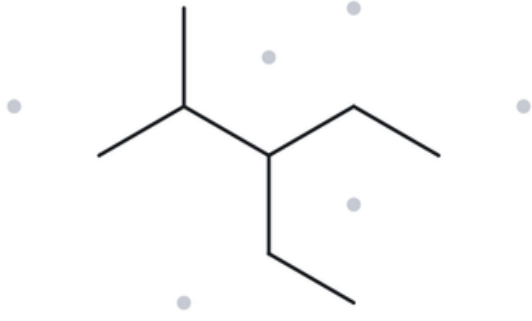
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

PARENT: PENTANE> A FIVE CARBON CHAIN

SUBSTITUENTS: ONE METHYL ON C2 FROM THE LEFT, ONE ETHYL ON C3

EXPECTED FORMULA: 5+3 = 8C, ONE CHAIN, TWO SUBSTITUENTS> C8H18

NAME DECODED



BOND-LINE STRUCTURE (THE ALKANE)

COUNTED FROM THE DRAWING: C8H18

ETHYL ON C3, METHYL ON C2

THEREFORE CORRECT!

FORMULA CHECK

RETURNED BY THE ALKANE: 3-ETHYL, 2-METHYLPENTANE

SAME AS GIVEN NAME!

BACK-CHECK

CARD B05

MOLECULAR FORMULA:

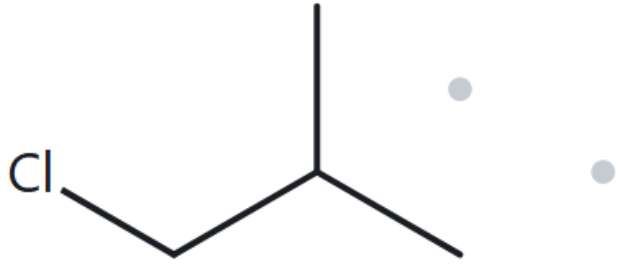
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

PARENT: PROPANE> A THREE CARBON CHAIN

SUBSTITUENTS: ONE CHLORINE ON C1 FROM THE LEFT, ONE METHYL ON C2

EXPECTED FORMULA: 3+1 = 4C, ONE CHAIN, TWO SUBSTITUENTS, ONE H IS REPLACED BY CHLORINE> C4H9CL

NAME DECODED



BOND-LINE STRUCTURE (THE ALKANE)

COUNTED FROM THE DRAWING: C4H9CL

CHLORINE ON C1, METHYL ON C2

THEREFORE CORRECT!

FORMULA CHECK

RETURNED BY THE ALKANE: 1-CHLORO-2-METHYLPROPANE

SAME AS GIVEN NAME!

BACK-CHECK

CARD B06

MOLECULAR FORMULA:

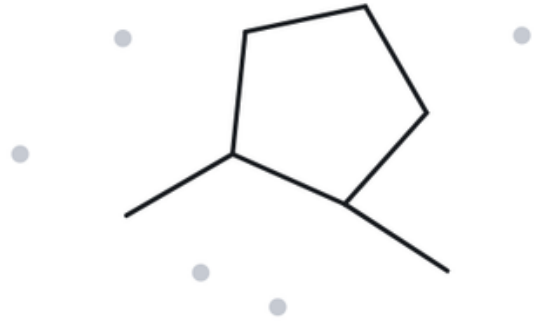
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

PARENT: CYCLOPENTANE> A 5 CARBON RING

SUBSTITUENTS: TWO METHYL, ONE ON C1, ONE ON C2, THEREFORE DIMETHYL

EXPECTED FORMULA: 5+2 = 7C, ONE RING, TWO SUBSTITUENTS>C7H14

NAME DECODED



BOND-LINE STRUCTURE (THE ALKANE)

COUNTED FROM THE DRAWING: C7H14

DIMETHYL, PRESENT IN C1, C2

THEREFORE CORRECT!

FORMULA CHECK

RETURNED BY THE ALKANE: 1,2-DIMETHYLCYCLOPENTANE

SAME AS GIVEN NAME!

BACK-CHECK

CARD B07

MOLECULAR FORMULA:

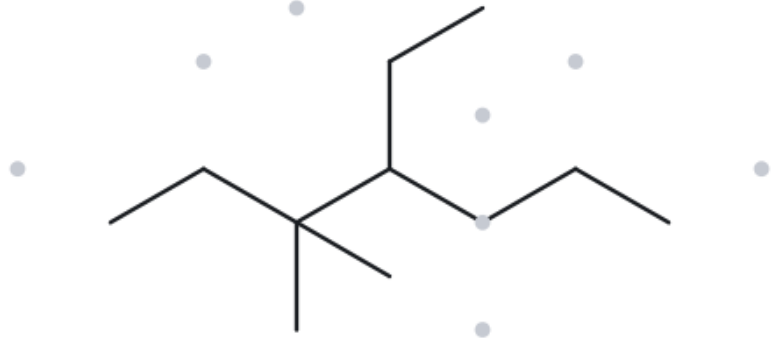
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

PARENT: HEPTANE> A SEVEN CARBON CHAIN

SUBSTITUENTS: TWO METHYL ON C3, THEREFORE DIMETHYL, ONE ETHYL ON C4

EXPECTED FORMULA: 7+4 = 11C, ONE CHAIN, THREE SUBSTITUENTS>C11H22

NAME DECODED



BOND-LINE STRUCTURE (THE ALKANE)

COUNTED FROM THE DRAWING: C11H22

DIMETHYL PRESENT IN C3, C2, ETHYL ON C4

THEREFORE CORRECT!

FORMULA CHECK

RETURNED BY THE ALKANE: 4-ETHYL-3,3-DIMETHYLHEPTANE

SAME AS GIVEN NAME!

BACK-CHECK

CARD B08

MOLECULAR FORMULA:

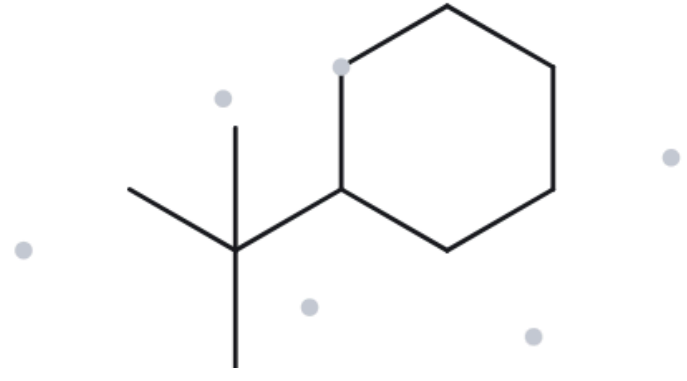
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

PARENT: CYCLOHEXANE> A SIX CARBON RING

SUBSTITUENTS: A TERT-BUTYL ON C1

EXPECTED FORMULA: 6+4 = 10C, ONE RING, ONE SUBSTITUENT>C10H20

NAME DECODED



BOND-LINE STRUCTURE (THE ALKANE)

COUNTED FROM THE DRAWING: C11H22

TERT-BUTYL PRESENT IN C1

THEREFORE CORRECT!

FORMULA CHECK

RETURNED BY THE ALKANE: (1,1-DIMETHYLETHYL)CYCLOHEXANE

(1,1-DIMETHYLETHYL) IS ALSO A NAME FOR TERT-BUTY, LIKE T-BUTYL, THEREFORE CORRECT!

BACK-CHECK

CARD B09

MOLECULAR FORMULA:

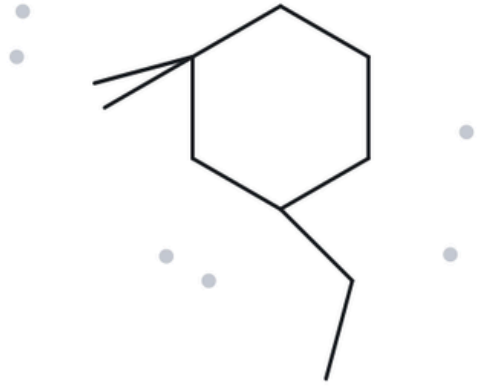
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

PARENT: CYCLOHEXANE> A SIX CARBON RING

SUBSTITUENTS: DIMETHYL ON C1, ETHYL ON C3 FROM THE LEFT

EXPECTED FORMULA: 6+4 = 10C, ONE RING, TWO SUBSTITUENTS>C10H20

NAME DECODED



BOND-LINE STRUCTURE (THE ALKANE)

COUNTED FROM THE DRAWING: C10H20

DIMETHYL ON C1, ETHYL ON C3

THEREFORE CORRECT!

FORMULA CHECK

RETURNED BY THE ALKANE: 3-ETHYL-1,1-DIMETHYLCYCLOHEXANE

THEREFORE CORRECT!

BACK-CHECK

CARD B10

MOLECULAR FORMULA:

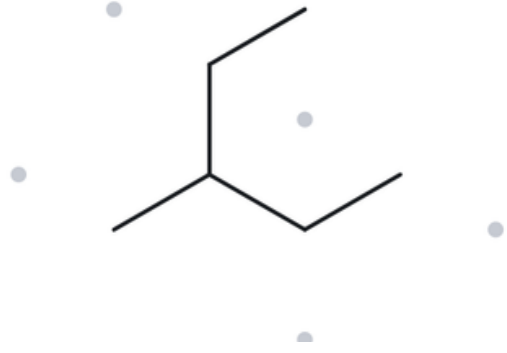
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

PARENT: THE PARENT IS NOT BUTANE, INFACT.... IT IS PENTANE!!! THEREFORE A 5 CARBON CHAIN

SUBSTITUENTS: THE SUBSTITUENT IS NOT ETHYL ON C2.... IT IS INFACT... METHYL ON C3!!!!

EXPECTED FORMULA: 5+1 = 6C, ONE CHAIN, ONE SUBSTITUENT>C6H14

NAME DECODED



BOND-LINE STRUCTURE (THE ALKANE)

COUNTED FROM THE DRAWING: C6H14

METHYL ON C3, COUNTED FROM THE LEFT

THEREFORE CORRECT!

FORMULA CHECK

RETURNED BY THE ALKANE: 3-METHYLPENTANE

THEREFORE, IT IS NOT 2-ETHYLBUTANE,BUT 30METHYLPENTANE!

BACK-CHECK